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Comparison of Sensors Performance for the Development of Wrist Pulse Acquisition System

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Abstract— The radial artery pulse signals have been the reliable and easy means of diagnosing a disease since ancient time, not only in India (Ayurveda) but also in many parts of the globe. This pulse diagnosis technique commonly known as the “*Nadi Pariksha*” has been recognized not only for monitoring the health status of patients but also for identifying the root cause of the disorder. However, it requires experience and expertise to learn, as a result, this highly promising technique is disappearing rapidly in the modern healthcare system. There is an acute need to save this proven and versatile art of investigating diseases based on very fundamentals of the biological system. In the past, research has been carried out for the computational pulse diagnosis, where sensors are used to acquire the *nadi* pulse wave for signal processing. Reliable acquisition of the pulse signal is an essential requirement for pulse based health monitoring and sensor's performance greatly affects the faithfulness of the signal to be processed. Therefore, in this work, the performance comparison of different pressure sensors is carried out for the design and development of wrist pulse acquisition system. The pressure sensors, used for pulse signal measurement can be grouped into two major categories i.e. piezoelectric and piezoresistive sensor. Different measurement constraint exists with both the sensor are discussed.

Keywords— *Pulse diagnosis, biomedical-instrumentation, sensors, data acquisition.*

I. INTRODUCTION

The *Nadi Pariksha* (Pulse Diagnosis) is the most important and versatile diagnostic methodology in India, which is also prevalent in many other countries as a part of their traditional medicine, e.g. China, Japan, Korea, etc. It is art and science of detecting the existing status of a person's body, mind, soul and spirit by examining the *Nadi* (Pulse) which is the rhythm/beat of the modulation of blood flow due to contraction and expansion of blood vessels, which can be felt at many locations of a human body. The pulse, although driven by the heart, is transmitted by blood flow through arteries and carries nutrients to the cellular level. There is an incessant flow of communication between cells, which is intelligence [1]. Pulse resembles as the music of the body's symphony, which is not a monotone (e.g., typically 70-80 times per minute) but is rich in harmonics. The frequency components (order) and their amplitude in the harmonics of the pulse contain most, if not all, of the information on the body health [2]. Thus, it is affected by whole activity of human body and its organs, like, the condition of the heart, nerves, muscles, skin, arterial walls,

blood parameters (volume, contents, viscosity, pressure, and velocity), etc. and govern functions of the body in normal conditions. Its different pattern can be strongly correlated with various abnormal conditions of the body as compared to its normal behavior/pattern, which enables us to use it for clinical diagnosis purpose.

This highly promising technique (Pulse Diagnosis) requires experience and expertise to learn, but in the modern era with many advanced healthcare instruments and technologies, there are very few experts who can practice this art effectively. In last few decades, many inventions have been reported, which try to save this proven and versatile art that appeared to be diminishing with significant reduction in the experts of this art.

In the past years, various sensors and systems have developed for accurate acquisition of pulse signals from the radial artery for computational pulse diagnosis. Lee *et al.* [3] used condenser microphone as a transducer to convert pulse on the wrist into electrical signals. Renevey *et al.* [4] proposed an infrared (IR) pulse detection system from the wrist for heart rate monitoring. Sorvoja *et al.* [5] developed a pressure sensor array based on electromechanical film (EMFi) for pulse signal acquisition. Joshi *et al.* [6] developed a pulse based diagnostic system using pressure sensor. Chu-Chang Tyan *et al.* [7] developed a non-invasive pulse monitoring system which uses a modified sensor specially designed to simultaneously measure the arterial diameter changed waveform and pulse pressure waveform. It uses a pressure sensor for recording the pressure pulse waveform and a strain gauge for recording the width of wrist pulses. Abhinav *et al.* [8] developed a system that monitors pressure variations from radial artery for pulse diagnosis using piezo based sensors. Yinghui Chen *et al.* [9] presented a modified auto-regressive model for computerized pulse diagnosis where a doppler ultrasound device is used for wrist pulse signal acquisition. Chung-Shing Hu *et al.* [10] present a pulse taking platform with a tactile array sensor. Wang *et al.* [11] presented a compound pressure signal acquisition system which overcomes the problems of sensor positioning, pressure adjustment resulting in a compound system for multichannel pulse signal acquisition. It integrates a pressure sensor with a photoelectric sensor to make a fusion sensor to feel pulse with different sensor arrays. Peng *et al.* [12] demonstrated flexible capacitive tactile sensors for real time pulse measurement. Wang *et al.* [13] integrated a pressure sensor with a photoelectric sensor to make a fusion sensor for pulse measurement and designed the multichannel sensor

arrays structure to acquire more pulse information. Wang *et al.* [14] combine a pressure sensor with a photoelectric sensor array to make a multichannel sensor fusion structure, that uses the optimal pulse signal processing methods and sensor fusion strategy for the feature extraction. Zuo *et al.* [15] presented the dependence and complementarity among three major types of sensors, i.e., pressure, photoelectric, and ultrasonic sensors, for pulse signal diagnosis. According to Zuo, pressure sensor is designed to measure the transmural pressure of the blood vessel, photoelectric sensor is designed to measure the blood volume of the blood vessel and ultrasonic sensor is employed to measure the blood velocity along the blood vessel of the wrist radial artery.

Ayurveda is one of the traditional system of medicine. Traditional practitioner feels the pulse signal of radial artery on the wrist at three specific position using three figure tips. Accurate and reliable measurement of pulse signal is crucial for computerized pulse diagnostic system. Although a number of sensors and systems have been developed for acquiring pressure pulse signal, it is essential to study the characteristics and properties of pulse signal obtained by these sensors. The following are the main objectives of this paper:

- To compare different pressure sensors performance for the design and development of pulse signal acquisition system.
- To study the effect of external pressure, position of sensors over radial artery, stiffness of hand on pulse acquisition has been discussed.

II. FRAMEWORK FOR EXAMINING AND ACQUIRING PULSE SIGNALS

There are several points in the human body to read the pulse. Out of them radial artery of the wrist is the most convenient one to read the pulse due to easy access, better hold. Usually, *Nadi Pariksha* is performed by placing three fingers of the physician to gently touch the skin over the radial artery to examine the pulse. Typically, the index finger is placed nearest to the thumb of the patient and the other two fingers, namely the middle finger and the ring finger are placed next to it as shown in Fig. 1. There should be little gap between the fingers so that the throbbing under each finger can be felt distinctly. During the examination, the examiner also notes the general conduct and habits of the patient. The appetite, facial expression, nature of sleep, breathing pattern, the reaction toward the climatic conditions, strength, history of ailments, all these facts also provides crucial input in accurate and precise diagnosis using pulse.

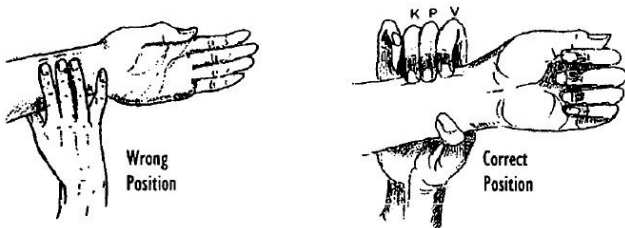


Fig. 1. Wrong and correct position of examining the pulse [1].

For the digitization of pulse diagnosis, it is very much necessary to first measure pressure pulse signal accurately and faithfully. Although a number of sensors and systems have been developed for acquiring pressure pulse signal, it is necessary to present the characteristics and properties of pulse wave form obtained by these sensors. We have discussed various aspect of pulse signal acquisition taking considerations of different parameter like external pressure, position of sensors over radial artery, stiffness of hand and so on. We have presented a comparison of pulse signals acquired by different sensors and the effect of above stated parameter on the acquired pulse signals.

Pulse signals are nothing but pressure waves, so pressure sensors are used to convert the mechanical pressure over the radial artery into electrical signals for digitization of *nadi pariksha*. A pressure sensor measures the pulse pressure wave when placed over the radial artery of the wrist. The pressure sensors described here for pulse acquisition fall into two broad categories namely piezoelectric and piezoresistive, whose working principle and pulse measurement mechanism are discussed broadly in the following sections.

III. PULSE MEASUREMENT TECHNIQUES

In this section, we describe the measurement techniques of two major types of sensors which are used to acquire the pulse signal from the radial artery. A typical pulse acquisition system consists of three major components, i.e., sensor, analog signal conditioner, and digitizer units (Fig. 2). Sensor is the most important building block of a pulse signal measurement systems, which first convert pulse signal electric signals, and then the analog signal conditioning circuit follows to amplify and filter the pulse signals, followed by digital circuit for the analog-to-digital (AD) conversion. The sampled pulse signal is stored in PC for further signal processing.

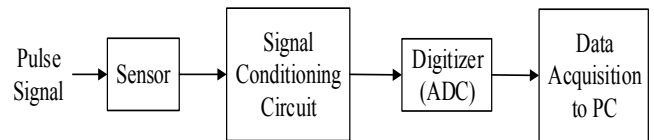


Fig. 2. Basic flowchart of pulse signal acquisition system.

Besides this, pulse acquisition system needs a structure for sensor placement to acquire the pulse more accurately at three different positions over radial artery as discussed earlier. Before acquiring the pulse signals, pulse beatings at the patient's wrist are first felt with three fingers and the rough positions of *Vata*, *Pitta*, and *Kapha* are searched and ensured for proper positioning of sensors over wrist radial artery.

The sensors described here for pulse acquisition fall into two broad categories namely piezoelectric and piezo resistive, whose working principle and pulse measurement methods are discussed broadly here.

A. Pulse measurement using piezoelectric sensor

Pulse signals are generated by the contraction and relaxation of heart during cardiac cycle. These signals are nothing but pressure wave which propagates throughout the

arterial system by blood. The transmitted pressure wave expands the arterial walls which can be best felt at the wrist radial artery. These pressure changes can be easily picked up by a piezoelectric diaphragm sensor, which generates equivalent electrical signal when subjected to change in pressure. The piezoelectric sensor uses the piezoelectric effect to measure changes in pressure, force or strain, acceleration. It is an active transducer i.e. if a dynamic force is applied to the crystal, it produces an electrical signal. Therefore, piezoelectric transducers are not good for static measurement. As pulse pressure waves are dynamic in nature, piezoelectric sensors are used to measure the pulse signals.

The Murata's 7BB-15-6LO piezoelectric sensor, shown in Fig. 3, has a piezoelectric diaphragm of diameter of 15 mm (equal to the size of human fingertip), making is a better choice for pulse pressure measurement. Because of its circular size it is quite easy to place the sensor over radial artery for pulse measurement application. It is having a resonant frequency of 6 kHz. Typically for signal conditioning of a piezoelectric sensor, there are two different approaches depending on the application. If the sensor is far from the signal conditioning circuit, charge amplifier circuit is used to convert the charge generated by the sensor into electrical voltage. If the sensor is close to the circuits, a simple voltage configuration might be sufficient.

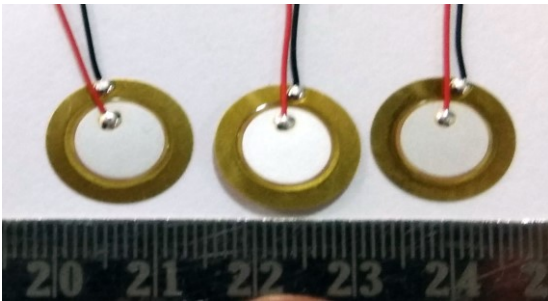


Fig. 3. Three piezoelectric diaphragm sensors.

Circuit Description:

Normal output voltages from piezoelectric sensors can vary from microvolts to tens of volts, and proper signal conditioning circuit is required for signal acquisition and processing. The first requirement is to amplify the sensor output to approximately 3 V to 5 V range for full scale. Two options are popular for this stage of signal conditioning, namely, a voltage mode amplifier (VMA) circuit, and a charge mode amplifier (CMA) circuit. VMA circuit is used when the amplifier is to be placed very close to the sensor and CMA circuit is used when the amplifier is remote to the sensor. In this work, a charge amplifier is used to convert the charge generated by the sensor into voltage. Fig. 4 shows the circuit schematic for charge mode based signal condition, where the output from charge amplifier is passed through low pass filter and gain stage before ADC conversion. The output signal from ADC is stored in PC for further processing. Resistor R_f drains the charge off capacitor C_f at a low rate to prevent the amplifier from drifting into saturation. Resistor R_f also provides a path for dc current.

The value of R_f and C_f set the low cut-off frequency of the amplifier. Resistor R_1 provides ESD protection. The biasing shown will put the output voltage at V_{ref} with no input, which helps in operating the circuit in a single supply voltage. The output will swing around this dc level.

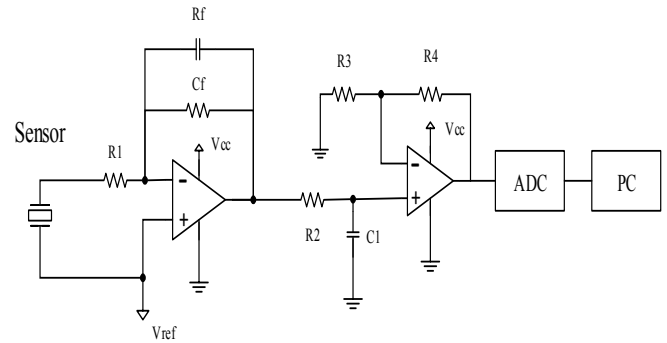


Fig. 4. Schematic of signal conditioning circuit for piezoelectric sensor.

The piezoelectric sensor has its own resonance frequency. In general, this frequency has to be filtered. Here, the piezoelectric sensor has a resonance frequency around 28 kHz. The pulse signal frequency varies from 0.5 Hz to 5 Hz. Low pass filter circuit is used also to remove the high frequency noise signal coupled with the pulse signal. The output signal from charge amplifier is passed through a first order low pass filter with cut-off frequency 40 Hz using resistor R_2 and C_1 is used to filter the high frequency component present in pulse signals. A gain stage of two times followed by an analog to digital converter (ADC) is used for data acquisition to the PC.

B. Pulse measurement using piezo resistive pressure sensor

The piezoresistive pressure sensor measures the pressure through a diaphragm formed on a silicon substrate, which bends with applied pressure. Because of the bending of the diaphragm, deformation occurs in the crystal lattice of the diaphragm. There are changes in the band structure of the piezo resistors attached to the diaphragm due to deformation, which leads to change in resistivity of the material.

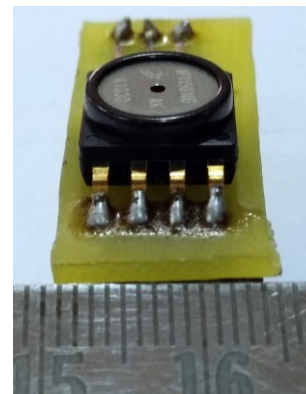


Fig. 5. MPX5010G6U- A silicon piezoresistive pressure sensor.

A silicon pressure sensor, MPX5010G6U, shown in Fig. 5 has internal temperature compensation and gain stage, which provides an accurate, high level analog output signal

proportional to the applied pressure. The analog output voltage is in the range of 0.2 to 4.7 V for a pressure range of 0 to 10 kPa (0 to 1.45 psi). The sensitivity of this MPX5010G6U pressure sensor is 450 mV/kPa.

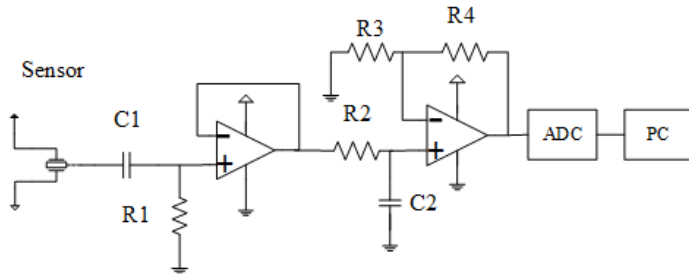


Fig. 6. Schematic of signal conditioning circuit for MPX5010G6U piezoresistive pressure sensor.

Circuit Description:

As the output voltage of sensor is amplified and in the range of 0 to 5 V, it can be directly fed into a microcontroller input for ADC conversion. However, it is having residual voltage. A high pass filter of 0.05 Hz is used to remove the DC component. Low pass filter of cut off frequency 40 Hz is used to remove the high frequency component from the pulse signal. Fig. 6 shows the circuit diagram for this sensor.

IV. RESULT AND DISCUSSION

In this experiment, we have used piezoelectric and piezoresistive sensor based system, which is mounted on the wrist to sense the pulse at three distinct locations on the radial artery, namely *Vata*, *Pitta*, and *Kapha*. The electrical signal generated by the sensor which is proportional to the pressure wave experienced on the radial artery, is then digitized using TI's EK TM4C123GXL microcontroller which is having a 12-bit analog to digital conversion (ADC). The data is captured at a sampling rate of 1000 Hz for a period of 2-3 minutes. The sampled data is stored in the PC using microcontroller which is having an USB interface with the personal computer. The *Vata* pulse signal of a healthy person acquired using piezoelectric and piezoresistive sensor based system at sampling frequency of 1 kHz is shown in Fig. 7.

A. Time Domain Characteristic of Obtained Pulse Signal

Before proceeding to the comparison of pulse signals acquired using piezoresistive and piezoelectric sensor based systems, validation of experimental results obtained is carried out by the time domain analysis of pulse signals. In the time domain, pulse signal waveform has unique characteristics such as a percussion wave (P), a tidal wave (T), valley (V) and a dircotic wave (D) as shown in Fig. 8. These characteristics have important role in classifying various patterns of pulse wave. The pulse wave acquired from a volunteer having these characteristics along with definite amplitude and time duration, indicates a good health condition. In addition to these features, in the frequency domain, the distribution of spectrum and the resonance of wrist pulse signals have recently been investigated. It is observed that a pulse waves are not monotonic in nature (means signal contains only a single

frequency component), but are rich in harmonics. Figure 3.3 shows a sample pulse waveform with time domain characteristics obtained using piezoelectric sensor based system.

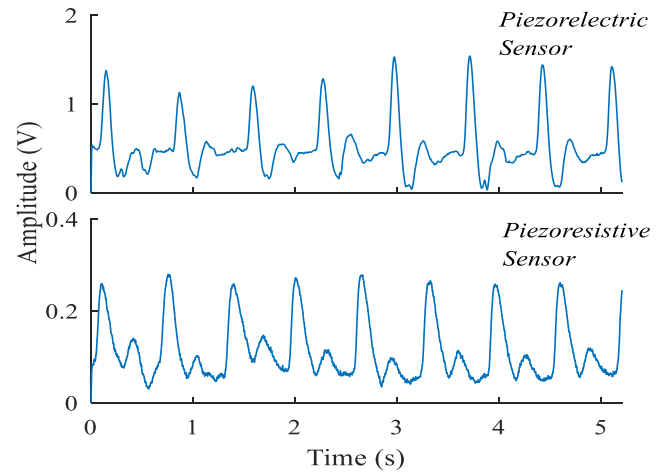


Fig. 7. Vata pulse signals of a healthy person.

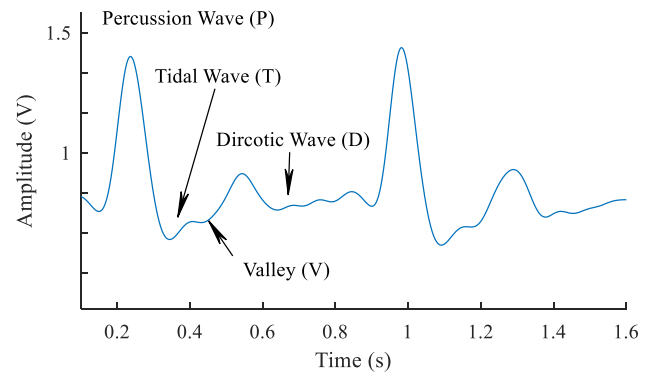


Fig. 8. A sample pulse signal.

B. Comparison of Pulse Signals Acquired Using Piezoelectric and Piezoresistive Sensor

In the case studies, we have compared the pulse signals acquired by using the piezoelectric and piezoresistive sensor based systems. We have presented the pulse signals acquired from four volunteers using both the sensors at three distinct locations on the radial artery for *Vata*, *Pitta* and *Kapha doshas*, respectively.

Female Volunteer - 1

Gender: Female Age: 35, Blood Group: B+, Weight: 55
Health Status: Normal

Fig. 9 shows the pulse signals of a healthy female volunteer acquired using piezoelectric and piezoresistive (MPX5010G6U) sensor. As compared to piezoelectric sensor

pulse signals, the pulse signals of MPX5010G6U are of same amplitude. Usually, the *Kapha* pulse signal is weak and it is difficult to acquire using piezoelectric sensor based system. But, the *Kapha* pulse acquired using MPX5010G6U sensor is having the same amplitude that of *Vata* and *Pitta* pulse. Also positioning of MPX5010G6U sensor over the radial artery for the acquisition of *Kapha* pulse is comparatively much easier than piezoelectric sensor.

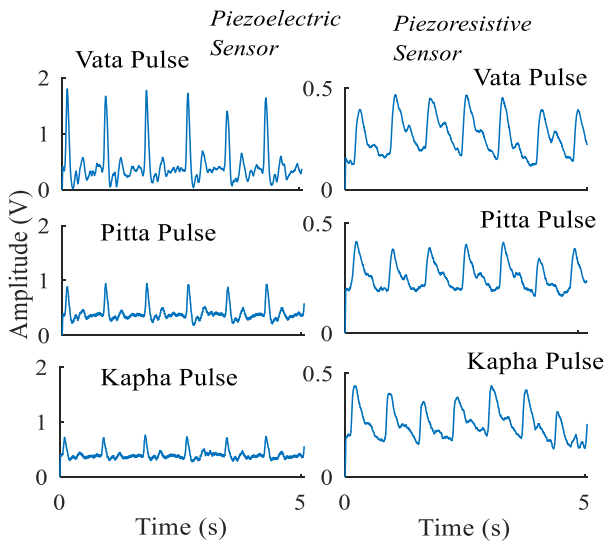


Fig. 9. Pulse signal of female volunteer - 1 obtained using piezoelectric and piezoresistive sensor based system.

Male Volunteer –1

Gender: Male, Age: 24, Blood Group: O+, Weight: 89
Health Status: Overweight.

Male Volunteer – 2

Gender: Male, Age: 25, Blood Group: O+, Weight: 68
Health Status: Stressed.

Figs. 10 and 11 show the pulse signals acquired using piezoelectric and piezoresistive (MPX5010G6U) sensor of two male volunteers. In the first case (Fig. 10), volunteer is overweight, whereas, in second case (Fig. 11) volunteer is having stressed mental condition. Here it is observed the pulse rate of the male volunteer is very high in second case. The pulse signals acquired using piezoelectric sensor based system contains more noise as compared the pulse signals of M5010G6U sensor based system.

Measurement errors are one of the most critical aspects of wrist pulse acquisition systems. Many a times pulse signal corrupted because of various parameters. Some of these includes: Positioning of Sensor, Motion Artefact's Due to Holding Structure, Base Line Shift, Electronics Noise, etc. Table-I presents the relative comparison of the two types of the sensors, which can be used for wrist pulse acquisition.

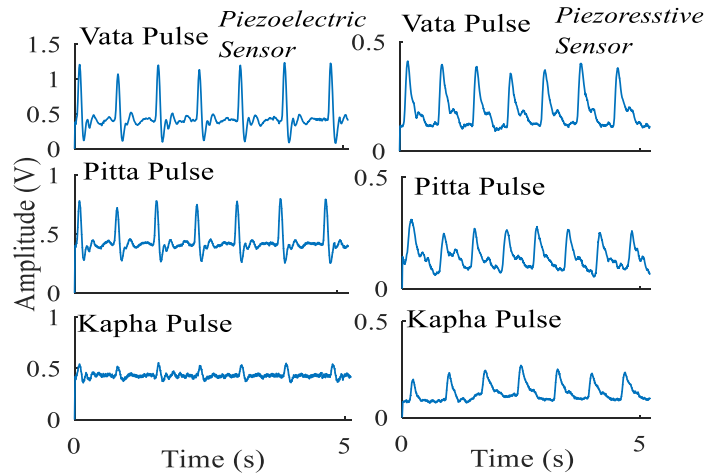


Fig. 10. Pulse signal of male volunteer - 1 obtained using piezoelectric and piezoresistive sensor based system.

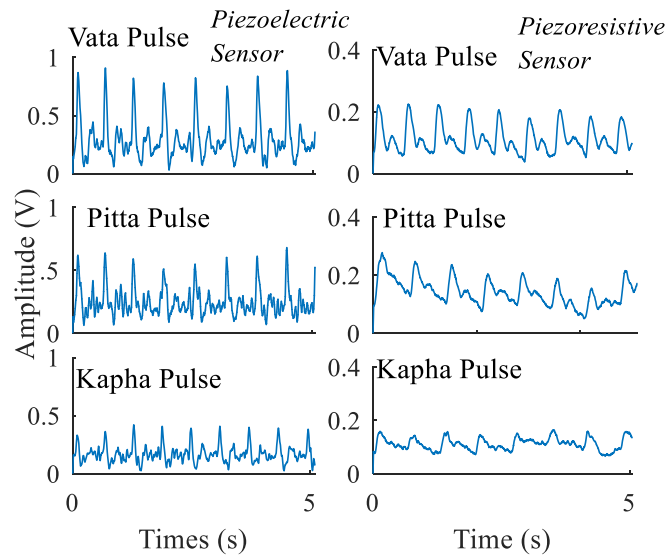


Fig. 11. Pulse signal of male volunteer - 2 obtained using piezoelectric and piezoresistive sensor based system.

Table-I Relative comparison of the sensors

Sensor	Advantage	Disadvantage	Result
Piezoelectric sensor	Circular diaphragm, easy to mount	Sensitive to small change in the wrist movement which introduce error	Produces accurate signals with proper positioning
MPX5010 G6U Piezoresistive sensor	Best for mounting. O/P voltage is temp compensated	High Pressure range reducing the resolution	Produces comparable better output (0.2 V to 1 V)

V. CONCLUSIONS

Pulse diagnosis is an important diagnostic tool for disease detection for thousands of years specifically in India and China as a part of their traditional diagnosis. To overcome the subjective nature of the pulse diagnosis, research has been carried out for the digitization of pulse diagnosis. In this work, wrist pulse signal acquisition system is designed using two types of pressure sensor namely piezoelectric and silicon piezoresistive pressure sensor. The calibration of sensors and specific parameters related with the pulse taking are discussed. An accurate, reliable, portable, non-invasive device is fabricated for pulse signal acquisition using piezoelectric sensors.

Case studies have been conducted to compare the pulse signals acquired using the above stated systems. The holding structure and positioning of sensor and their effect in the acquisition of pulse signal are discussed in detail. Different measurement errors during the acquisition of pulse signal are discussed. The strength and shape of the pulse waveform are discussed for the two-pressure sensor based systems. The experimental results show that, in terms of accuracy, sensitivity and positioning MPX5010G6U piezoresistive pressure sensor is superior to piezoelectric sensor.

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